

Dear PNNL hiring team,

I believe I am a strong candidate for your Undergrad Tech Intern - Research Computing Internship (PEARC), and more specifically, your Data Engineering track.

My current research in low rank, data integration methods deals with a large number of parameters, heavy simulation calculations, and even heavier estimation calculations. I have built a full Python object-oriented simulation framework that handles and stores these large computations. I am familiar with creating data scraping, cleaning, and modeling pipelines that feed into various models, including but not limited to: linear regression, LASSO, KNN, bagging, random forest, LDA, QDA, and Naive Bayes. And, unfortunately, I am familiar with slow run time speeds for parameter expensive machine learning methods. and have tried to outsource my computing to beyond my local machine.

While I believe I am well-aligned with the active requirements for the Data Engineering track and would benefit greatly from this opportunity, PNNL would receive a hard-working, motivated, and dedicated research intern who is well versed in machine learning and statistical computing. It would be a privilege to use PNNL's high performance computing and apply it to enhance algorithm run times, target large and dense datasets, and run tests on models with an excruciating number of parameters.

I appreciate your time reading this cover letter, and I hope that our paths meet in the future.

Ryan